

PLASMA DISPOSAL OF PLASTIC WASTE AND GENERATION OF SYNGAS FOR POWER GENERATION

BRIEF DESCRIPTION:

The rapid rate of worldwide urbanization has led to increasing plastic waste generation. This increase has resulted in a large amount of plastic waste, particularly plastic bags and PET bottles, being littered on land filled. To address this problem a new technology has been developed for volume reduction of the waste streams through plasma pyrolysis resulting in by-product syngas which is a clean fuel

Salient Features: The bricks and building material obtained by this process have

- Uniform gasification, Offers a single step treatment
- Complete destruction of plastic waste
- Reduce the need for landfill
- Safer disposal of plastic waste

Advantage:

- Energy recovery system- contributing saving of resources
- Provides volume reduction of the waste streams
- The extreme high temp. of plasma kill stable bacteria
- Provide electricity to power the process
- Supply excess electricity for sale
- Formation of complex molecules , such as dioxins, is prevented
- Reduce the need for landfill
- Supply excess electricity for sale

Scale of development: Commercialized

